SO-ARM101

Participant Guide

Step-by-step guide to set up, calibrate, and operate your robot arm.
All commands included for Windows, macOS, and Linux.

TURTLE Robotics — SO-ARM101 Workshop Series

STEP 01

Install Software

Windows only: Use Miniforge Prompt (Start → Miniforge3), not PowerShell or Command Prompt.

Windows (Miniforge Prompt)

```
conda create -n lerobot python=3.10 -y
conda activate lerobot
conda install pytorch torchvision torchaudio cpuonly -c pytorch -y
cd C:\
git clone https://github.com/huggingface/lerobot.git
cd lerobot
pip install -e ".[feetech]"
python -c "import lerobot; print(lerobot.__version__)"
```

macOS

```
brew install miniforge
conda init zsh
# Restart terminal, then:
conda create -n lerobot python=3.10 -y
conda activate lerobot
cd ~
git clone https://github.com/huggingface/lerobot.git
cd lerobot
pip install -e ".[feetech]"
```

Linux

```
sudo apt update && sudo apt install -y git python3-pip ffmpeg
sudo usermod -aG dialout $USER
# Log out and log back in, then:
conda create -n lerobot python=3.10 -y
conda activate lerobot
cd ~
git clone https://github.com/huggingface/lerobot.git
cd lerobot
pip install -e ".[feetech]"
```

STEP 02

Find Your USB Port

Plug in your robot arm USB cable. Find the port name — you need it for every command below.

Windows (Miniforge Prompt)

```
# Device Manager -> Ports (COM & LPT)
# Look for: USB Serial Device (COM4)
# Or run:
Get-PnpDevice -Class Ports | Where-Object Status -eq OK
# Write down your port numbers! e.g. COM4 (follower), COM5 (leader)
```

macOS

```
ls /dev/tty.usbserial*
# Example: /dev/tty.usbserial-58FE017843
```

Linux

```
ls /dev/ttyACM*
# Follower: /dev/ttyACM0    Leader: /dev/ttyACM1
```

If nothing appears: check the USB cable, confirm 12V power is connected, try a different USB port.

STEP 03

Calibrate Your Arm

Calibration maps each joint's full range of motion. Do this once per arm. When prompted, slowly move each joint to its maximum and minimum position, then press Enter.

Windows (Miniforge Prompt)

```
cd C:\lerobot
conda activate lerobot
# Follower arm (replace COM4 with your port):
lerobot-calibrate --robot.type=so101_follower --robot.port=COM4 --robot.id=so_follower
# Leader arm (replace COM5):
lerobot-calibrate --robot.type=so101_leader --robot.port=COM5 --robot.id=so_leader
```

macOS

```
cd ~/lerobot && conda activate lerobot
lerobot-calibrate --robot.type=so101_follower \
  --robot.port=/dev/tty.usbserial-YOURPORT --robot.id=so_follower
lerobot-calibrate --robot.type=so101_leader \
  --robot.port=/dev/tty.usbserial-YOURPORT2 --robot.id=so_leader
```

Linux

```
cd ~/lerobot && conda activate lerobot
lerobot-calibrate --robot.type=so101_follower \
  --robot.port=/dev/ttyACM0 --robot.id=so_follower
lerobot-calibrate --robot.type=so101_leader \
  --robot.port=/dev/ttyACM1 --robot.id=so_leader
```

Move joints slowly all the way to each limit. Incomplete calibration causes jerky or stuck movement.

STEP 04

Teleoperation

Move the leader arm by hand — the follower mirrors it in real time. Press Ctrl+C to stop.

Windows (Miniforge Prompt)

```
cd C:\lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot teleoperate \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=COM4 \
  leader_arms.main.port=COM5
```

macOS

```
cd ~/lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot teleoperate \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/tty.usbserial-FOLLOWER \
  leader_arms.main.port=/dev/tty.usbserial-LEADER
```

Linux

```
cd ~/lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot teleoperate \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/ttyACM0 \
  leader_arms.main.port=/dev/ttyACM1
```

All 6 joints should move smoothly. If a joint goes the wrong way, ask your instructor about adjusting drive_mode in the calibration file.

STEP 05

Record Demonstrations

Record episodes of your arm completing a task. Aim for 50+ episodes of consistent, smooth demonstrations. Move slowly, start from the same position every time, and complete the full task each episode.

Log in to HuggingFace first:

```
huggingface-cli login
```

Windows (Miniforge Prompt)

```
cd C:\lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot record \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=COM4 leader_arms.main.port=COM5 \
  --fps 30 --repo-id YOUR_USERNAME/my_dataset \
  --tags so101 --warmup-time-s 2 \
  --episode-time-s 30 --reset-time-s 5 --num-episodes 50
```

macOS

```
cd ~/lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot record \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/tty.usbserial-FOLLOWER \
  leader_arms.main.port=/dev/tty.usbserial-LEADER \
  --fps 30 --repo-id YOUR_USERNAME/my_dataset \
  --tags so101 --warmup-time-s 2 \
  --episode-time-s 30 --reset-time-s 5 --num-episodes 50
```

Linux

```
cd ~/lerobot && conda activate lerobot
python -m lerobot.scripts.control_robot record \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/ttyACM0 leader_arms.main.port=/dev/ttyACM1 \
  --fps 30 --repo-id YOUR_USERNAME/my_dataset \
  --tags so101 --warmup-time-s 2 \
  --episode-time-s 30 --reset-time-s 5 --num-episodes 50
```

STEP 06

Train a Policy

Train a neural network to perform your task autonomously from the recorded demonstrations. Training takes 1-2 hours with a GPU, or 8-12 hours on CPU. Your instructor may run this overnight.

All platforms (from inside lerobot folder, env activated):

```
python -m lerobot.scripts.train \
  --policy-path lerobot/configs/policy/act.yaml \
  --dataset-repo-id YOUR_USERNAME/my_dataset \
  --output-dir outputs/train/act_so101 \
  --training.num-steps 80000 \
  --wandb.enable false
```

Run the Trained Policy

Windows (Miniforge Prompt)

```
python -m lerobot.scripts.control_robot eval \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" robot.port=COM4 \
  --policy-path outputs/train/act_so101/checkpoints/last/pretrained_model \
  --num-eval-episodes 5
```

macOS

```
python -m lerobot.scripts.control_robot eval \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/tty.usbserial-FOLLOWER \
  --policy-path outputs/train/act_so101/checkpoints/last/pretrained_model \
  --num-eval-episodes 5
```

Linux

```
python -m lerobot.scripts.control_robot eval \
  --robot-path lerobot/configs/robot/so101_follower.yaml \
  --robot-overrides "~cameras" \
  robot.port=/dev/ttyACM0 \
  --policy-path outputs/train/act_so101/checkpoints/last/pretrained_model \
  --num-eval-episodes 5
```